DFW Housing Market Analysis

Executive Summary

Prepared For: Executive Leadership, Real Estate Brokerage
Prepared By: Sam Corbin
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Analysis Period: May 2016 – May 2026
Tools Used: Google BigQuery, SQL, Tableau, Microsoft Excel, GitHub

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Executive Summary

This report presents a comprehensive analysis of residential housing market trends across the Dallas–Fort Worth (DFW) metropolitan area over a ten-year period, from May 2016 through May 2026. Using the Zillow Home Value Index (ZHVI), the project evaluated home values, historical appreciation, and affordability to identify markets that provide the strongest opportunities for prospective homebuyers.

The objective of the analysis was to determine which DFW cities offer the best balance of affordability, long-term appreciation, and overall buyer opportunity. To support this objective, housing data was extracted, cleaned, analyzed using SQL in Google BigQuery, and transformed into interactive Tableau dashboards designed for business decision-making.

The project resulted in four executive dashboards that provide insight into:

- Current housing market conditions
- Historical home value appreciation
- Buyer Opportunity Index rankings
- Executive Market Summary

Together, these dashboards enable stakeholders to compare cities and counties, identify emerging housing markets, and make data-driven recommendations for buyers.

Business Problem

Homebuyers frequently face the challenge of balancing affordability with long-term investment potential. Lower-priced markets may not generate significant appreciation, while rapidly appreciating markets often become less affordable over time.

Without reliable market intelligence, buyers risk either overpaying in highly competitive markets or missing opportunities in growing communities that offer stronger long-term value.

This project addresses that challenge by providing an objective, data-driven evaluation of housing markets across the Dallas–Fort Worth metroplex.

Project Objectives

The primary objectives of this analysis were to:

- Analyze current home values across DFW cities.
- Measure ten-year home value appreciation.
- Compare housing trends across counties.
- Identify cities that outperform regional averages.
- Develop a Buyer Opportunity Index that combines affordability and appreciation into a single comparative score.
- Present findings through interactive dashboards suitable for executive decision-making.

Data Source

The analysis is based on the Zillow Home Value Index (ZHVI), a widely recognized measure of typical home values across U.S. housing markets.

Dataset Characteristics

- Geographic Scope: Dallas–Fort Worth Metropolitan Area
- Analysis Level: City and County
- Time Period: May 2016 – May 2026
- Source: Zillow Home Value Index (ZHVI)

Five analytical datasets were developed during the project:

- latest_home_values.csv
- city_appreciation.csv

- county_appreciation.csv
- buyer_opportunity_index.csv
- county_performance.csv

These datasets support the SQL analysis and four interactive Tableau dashboards, covering current home values, historical appreciation, buyer opportunity, and overall county performance.

Methodology

The project followed a structured analytics workflow aligned with industry best practices.

1. Data Collection

Housing market data was obtained from the Zillow Home Value Index and imported into Google BigQuery for analysis.

2. Data Preparation

Data was reviewed for completeness and consistency before analysis. Relevant fields were selected and organized into analytical datasets supporting the project objectives.

3. SQL Analysis

SQL queries were developed to calculate:

- Current home values
- Average home values
- Highest and lowest home values
- City appreciation percentages
- County appreciation averages
- Buyer Opportunity Scores
- City and county rankings

4. Data Visualization

Interactive Tableau dashboards were created to communicate findings using:

- KPI scorecards

- Bar charts
- Geographic bubble maps
- Interactive dashboard actions
- Cross-filtering between visualizations

5. Business Interpretation

Analytical results were translated into actionable recommendations that support homebuyer decision-making and market evaluation.

Key Findings

The analysis produced several significant insights regarding the DFW housing market.

County-level findings from dashboard 4:

- **Overall leader:** Tarrant County, with a County Performance Score of 4.33 and an Overall Rank of 1.
- **Highest average home value:** Tarrant County, approximately $537,450.35.
- **Highest 10-year appreciation:** Hunt County, approximately 119.75%.
- **Lowest average Buyer Opportunity Score:** Van Zandt County, 50.00—the strongest result under the Buyer Opportunity scoring model.
- **Highest average Buyer Opportunity Score:** Rockwall County, 250.00—the weakest result under the Buyer Opportunity scoring model.

(These findings represent different performance measures. The county leading one metric does not necessarily lead the others. The County Performance Score combines three component rankings, with lower scores indicating stronger overall performance under the defined methodology.)

Strong Long-Term Growth

Many DFW cities experienced substantial home value appreciation during the ten-year analysis period, reflecting sustained demand and continued regional economic expansion.

Significant Market Variation

Home values varied considerably across cities and counties, demonstrating that buyers have access to a broad range of affordability options depending on location.

Buyer Opportunity Index

By combining affordability rankings with appreciation rankings, the Buyer Opportunity Index highlighted cities that provide strong value while maintaining favorable long-term growth potential.

County-Level Differences

Average home values and appreciation rates differed across counties, emphasizing the importance of evaluating markets beyond individual home prices.

Interactive Decision Support

The Tableau dashboards allow users to quickly compare markets, identify trends, and explore opportunities through interactive filtering and geographic visualization.

Business Recommendations

Based on the findings, the following recommendations are provided for real estate professionals and prospective buyers.

Prioritize High Opportunity Markets

Focus buyer searches on cities that rank highly within the Buyer Opportunity Index, as these markets provide a favorable combination of affordability and historical appreciation.

Evaluate Markets Beyond Price Alone

Current home price should not be the sole factor when evaluating purchasing decisions. Historical appreciation trends provide valuable insight into long-term investment potential.

Leverage County Comparisons

County-level trends can reveal broader market dynamics that may influence future growth and investment opportunities.

Incorporate Data into Client Consultations

Real estate professionals should integrate market analytics into buyer consultations to provide objective recommendations supported by historical market performance.

Continue Monitoring Market Trends

Housing markets evolve continuously. Regular updates to the analysis will ensure recommendations remain aligned with current market conditions.

Conclusion

This project demonstrates the complete data analytics lifecycle by transforming raw housing market data into actionable business intelligence.

Using SQL, Google BigQuery, Tableau, and Excel, the analysis delivers a professional decision-support solution that enables stakeholders to evaluate affordability, appreciation, and buyer opportunity across the Dallas–Fort Worth housing market.

The resulting dashboards provide an intuitive and interactive framework for exploring housing trends while illustrating the practical value of data analytics within the real estate industry.

Beyond answering the project's central business question, this analysis showcases the technical and analytical skills required of a modern data analyst, including data preparation, SQL querying, business intelligence development, data visualization, and executive storytelling.